

DiskVision: AI-Powered Disk Analysis and Storage Optimization

Graduation Project Poster (FCAI) - Designed with the same visual identity as the project website

Under Supervision of: Dr. Amr Hassanain | Course: Work-based Professional Project (Level 2) | Project Type: Desktop Application for Windows

Name of the Students

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Project Title

DiskVision: An Intelligent Disk Cleaner, Analyzer, and AI Advisor for Windows Users.

Abstract

DiskVision is a Windows desktop tool that scans storage, detects duplicates, finds temporary files, visualizes disk usage, and provides AI recommendations for safe cleanup. The goal is to help users reclaim space confidently while protecting important files.

Objectives

- Automate disk analysis and safe cleanup suggestions.
- Reduce storage waste from duplicates and temporary data.
- Provide clear visual and AI-assisted decision support.
- Improve system responsiveness and user experience.

Methodology

- Recursive file scanning and indexed search.
- Duplicate detection using hash-based comparison.
- Treemap visualization for folder size distribution.
- AI classification (Safe / Caution / Dangerous).

Significance of the Project

The project combines multiple storage-management tools in one platform, reducing user confusion and minimizing risky deletion decisions through explainable AI recommendations.

Results and Analysis

- Accurate identification of large and duplicate files.
- Measurable freed space after temp/duplicate cleanup.
- Stable performance on large folders with indexed search.
- Clear AI feedback with fallback behavior when server is unavailable.

Recommendations

- Add cloud storage integration (Drive/OneDrive).
- Support scheduled automatic cleanup rules.
- Improve duplicate detection for near-identical media.
- Extend support to macOS and Linux in future versions.

References

- CCleaner Official Documentation
- WinDirStat Documentation
- Duplicate Cleaner Technical Notes
- TreeSize Product Documentation
- Everything Search Documentation